

Elier Garcia

AI ENGINEER | DATA & AI

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Spain

Professional Summary

I didn't start my career behind a keyboard. I started reading electrical blueprints and negotiating with suppliers on active construction sites. After two years managing large-scale electrical infrastructure for Microsoft and BMW Group, I pursued a Master's in Business Analytics & AI, graduating top of my class. That's when it clicked: the operational inefficiencies I had witnessed firsthand in the field could be systematically addressed through data engineering, AI, and workflow automation. So I started building the solutions.

Skills

- ▶ RAG & LLMs
- ▶ n8n · Workflow Automation
- ▶ Machine Learning
- ▶ GCP (Google Cloud Platform)
- ▶ Python
- ▶ Next.js · React · TypeScript
- ▶ ETL / Data Pipelines
- ▶ SQL
- ▶ Technical Leadership
- ▶ Project Management
- ▶ Technical Communication

Web & Social

LinkedIn:
<https://www.linkedin.com/in/eliervg/>
Personal site:
<https://egarciav99.github.io>

Languages

Spanish — Native
English — B2
German — A2

Certifications

Top Graduate — Master's in Business Analytics & Artificial Intelligence

INESDI Business Techschool
Mar 2026 — Present

Work Experience

Full-Stack AI Developer

Freelance | Madrid | Aug 2025 — Present

Bocado AI — Personalized Nutrition SaaS · ~100 active users · Next.js 16 / Firebase / Gemini Flash

- Architected a real-time inference pipeline with progressive context enrichment prior to LLM invocation, injecting user nutritional profile, pantry state with urgency flags, interaction history, and feedback signals, served from a 5-minute TTL cache.
- Eliminated geographic hallucinations by inverting the standard generation flow: Google Places API pre-populates the prompt before inference, ensuring Gemini reasons over verified restaurant data rather than generating it. Operational cost: €0.50/month at real user scale.

CoverCraft — AI Cover Letter Generator · 4 languages · n8n / Gemini / Vanilla JS

- Designed and implemented a fully asynchronous end-to-end pipeline in n8n — webhook intake → PDF extraction → CV parser → LLM inference → frontend callback — decoupling generation from the client request/response cycle for non-blocking UX.
- Built a proprietary multilingual CV parser with section detection across ES/EN/FR/NL via text normalization, paired with ATS-oriented prompt engineering structured around four explicit-purpose paragraphs and a curated blacklist of generic adjectives.

PDF Tech Assistant — RAG System for Technical Manuals · React / Firebase Cloud Functions / Supabase pgvector

- Implemented a full RAG pipeline: PDF ingestion → hybrid semantic chunking → embedding via Gemini Embedding (768 dims) → top-8 retrieval in pgvector → generation with Gemini 2.0 Flash at temperature 0.1 for precision-critical technical responses.
- Deployed a two-layer caching strategy (Firestore-backed embedding cache + 1-hour query-hash cache) with p-limit concurrency control and exponential backoff to prevent API throttling on large-volume manuals.

Electrical Construction Site Supervisor

DEMEK S.A de C.V | Querétaro, México | Jan 2024 — Feb 2025

- Promoted at six months to lead the full electrical scope of a Microsoft Data Center, overseeing crews of up to 60 personnel across medium voltage systems, substations, generator sets, and pump room installations.
- Designed from scratch a materials traceability system, combining SAP consumption exports with comparative Excel tracking, for the 19,847 m² BMW Group plant expansion, achieving full oversight across all 8 project sections.
- Maintained zero supply interruptions over 14 cumulative months of execution, managing technical RFIs for low and medium voltage systems and grounding installations across industrial facilities.

Electrical Budget Analyst

DEMEK S.A DE C.V | Chihuahua, México | Jan 2023 — Jan 2024

- Owned the full estimating cycle: AutoCAD take-offs → Excel quantity surveys → OPUS cost modeling → supplier negotiation → client presentation → change order management during execution.
- Delivered 30+ proposals (up to USD 500K) with an average cost deviation below 3%, consistently within 48-hour turnaround windows. Terex Project (Nuevo León): sole proposal to reach final presentation stage, delivered directly in English to the client. DEMEK was awarded and executed the full plant.

Education

Master's Degree in Business Analytics & Artificial Intelligence

INESDI Business Techschool | Madrid, Spain | Jan 2026

- Specialized in ETL/ELT pipelines, RAG architectures, applied Machine Learning, and Power BI data visualization.

Bachelor's Degree in Electrical Engineering

Instituto Tecnológico de Chihuahua | Chihuahua, México | Jan 2023

- Major in Electrical Installations. Graduated with a GPA of 9.27/10.
- President of the Electrical Engineering Student Committee.